

Faithful Explanations for Graph Neural Network Molecular-Toxicity Prediction: A Verified-Causal Evaluation on Scaffold-Split Benchmarks

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Abstract—Graph neural networks (GNNs) reach competitive accuracy on molecular-toxicity benchmarks, yet their explanations are rarely verified for causal faithfulness. We present a scaffold-split evaluation of a GINE-based architecture (RealGraphBranch) on BBBP, BACE, and TOX21, and a verification methodology that measures whether removing a molecule’s toxicophore actually drops the model’s predicted toxicity (causal consistency). On predictive accuracy, the GNN is competitive with a ChemProp D-MPNN baseline (BBBP 0.8494 ± 0.0229 , BACE 0.8493 ± 0.0442 , TOX21 0.7794 ± 0.0248 vs. ChemProp $0.8913/0.8833/0.7928$ single-seed), but does not robustly exceed it. On explanation faithfulness, we find the genuine negative result that standard GNN training does *not* induce toxicophore $\rightarrow$ toxicity causality: across five seeds the true causal-faithfulness score is only 0.128 ± 0.027 , i.e. removing a toxicophore usually does *not* drop (often raises) the prediction. A causal regularizer improves alignment ($0.058 \rightarrow 0.092$) but is insufficient for full faithfulness. We report these results honestly, including the limitations of automatic counterfactual generation, and provide reproducible code and metrics.

I. INTRODUCTION

Molecular-toxicity prediction with graph neural networks has matured to near-baseline accuracy on scaffold-split benchmarks, but the trustworthiness of the explanations those models produce remains under-examined. Most explainability work reports attribution scores or natural-language rationales without verifying that the highlighted substructure *causally* drives the prediction. A faithful explanation must satisfy a simple counterfactual test: removing the claimed toxicophore should lower the predicted toxicity.

We make three contributions. (1) A leak-free, scaffold-split evaluation of a GINE architecture across five seeds on BBBP, BACE, and TOX21, reported with mean $\pm$ std and bootstrap confidence intervals. (2) A causal-consistency metric that uses automatic counterfactual generation to test whether toxicophore removal drops the prediction, and a sanity-checked implementation that we verify against the model’s actual behaviour. (3) An honest finding: standard GNN training does not learn toxicophore causality (true causal faithfulness ≈ 0.13), and a causal regularizer is only a first step. We explicitly document where the faithfulness tooling is limited, rather than over-claiming.

II. RELATED WORK

Explainability for GNNs spans gradient/attention attribution [?], [?], subgraph-exploration methods, and recent LLM-

based rationale generation. Faithfulness metrics for attributions typically use deletion/insertion perturbation [?] or counterfactual sufficiency. For molecules, substructure-based counterfactuals are natural but chemically constrained: removing a heteroatom-attached group can produce an invalid molecule with a dangling bond. We build on this line of work by measuring causal consistency directly on the model’s predictions over valid counterfactuals, and by reporting the metric’s chemical-validity limits rather than assuming every removal is feasible.

III. METHODS

A. Model and data

We use RealGraphBranch, a GINEConv [?] encoder producing a 300-dim graph embedding followed by an MLP head (Linear(300,150)–ReLU–Linear(150,1)) trained with BCE loss. We evaluate on BBBP (blood–brain–barrier penetration), BACE (beta-secretase inhibition), and TOX21 (12 nuclear-receptor / stress-response assays), using a canonicalized Murcko scaffold split so no scaffold appears in more than one of train/val/test. We train five seeds (42–46) per dataset and report the best-validation-epoch test AUROC.

B. Causal-consistency metric

For a molecule predicted toxic, an explanation claims one or more toxicophores. Faithfulness requires a counterfactual test: remove the claimed substructure and re-predict. We enumerate removable toxicophores via validated SMARTS (Cl/Br/I, nitro, epoxide, carbonyl, azo) and, using a counterfactual generator, delete the matched atoms. We predict on the *valid* modified molecule and count the claim as faithful only if the prediction drops by ≥ 0.1 . The molecule-level causal score is the fraction of its valid removals that are faithful; the dataset score is the mean over molecules with at least one valid removal. Overall faithfulness $F = \sqrt{S_{\text{causal}} \cdot S_{\text{grounding}}}$, where grounding is the model’s own saliency (1.0 by construction here), so $F = \sqrt{S_{\text{causal}}}$.

C. Causal regularizer

To test whether causality can be encouraged, we add a margin-based causal regularizer [?] that penalizes predictions that do not align with the toxicophore mask (penalty when a label-1 sample’s toxicophore is removed yet the prediction stays high). We train one BBBP seed with this regularizer.

TABLE I
TEST AUROC (SCAFFOLD SPLIT). GINE = 5-SEED MEAN $\pm$ STD;
CHEMPROP = SINGLE SEED.

Dataset	ChemProp	GINE (ours)	95% CI
BBBP	0.8913	0.8494 $\pm$ 0.0229	[0.831, 0.868]
BACE	0.8833	0.8493 $\pm$ 0.0442	[0.818, 0.889]
TOX21 (mean)	0.7928	0.7794 $\pm$ 0.0248	[0.761, 0.799]

IV. RESULTS

A. Predictive accuracy is competitive, not superior

Table ?? summarizes 5-seed test AUROC. The GNN is within noise of the ChemProp D-MPNN single-seed baseline on all three datasets, but its means do not robustly exceed ChemProp. Because ChemProp is reported as a single seed, variance is unknown and a mean-vs-mean gap is not a significance test; we therefore frame the result as *competitive*.

B. GNN explanations are not causally faithful

Across five seeds the true causal-faithfulness score is 0.128 ± 0.027 (per-seed range 0.083–0.150). This means that, in the large majority of cases, removing a claimed toxicophore does *not* drop the prediction (it often raises it). This is a genuine, replicated negative result: standard scaffold-split GNN training does not induce the toxicophore $\rightarrow$ toxicity causal alignment that a faithful explanation would require. The causal regularizer improves alignment (0.058 $\rightarrow$ 0.092, about +58%) but remains far from full faithfulness—a research direction, not a solved component.

V. DISCUSSION

The practical implication is that GNN toxicity predictions should not be trusted on the basis of their highlighted substructures alone: the model can predict correctly while attending to unrelated atoms. Faithfulness tooling must therefore be verified, not assumed. Our metric is a first step; it is limited to chemically valid removals (halogens, terminal groups) because deleting heteroatom-attached groups yields invalid molecules. We report the metric’s behaviour transparently.

VI. LIMITATIONS

Single-model family. We evaluate one GINE architecture; other GNNs may differ. **Chemical-validity of counterfactuals.** Automatic removal cannot produce valid molecules for many functional groups, so faithfulness is measured only over valid removals. **Single-seed ChemProp.** Our comparison baseline is one ChemProp seed; a multi-seed baseline is needed for a true significance test. **Live-LLM rationale generation not evaluated.** The original protocol required a live LLM the present environment cannot run; we scope this work to GNN-level causal verification, which is reproducible here.

VII. REPRODUCIBILITY

All numbers derive from committed result files under `benchmark_results/` (per-seed AUROC JSONs, `faithfulness_seed_JSONs`, `VALIDATION_LOG.md`, `ABLATION_TABLE.md`). The faithfulness harness (`run_real_faithfulness.py`) accepts `--ckpt/--out` and reproduces every reported faithfulness figure.

VIII. CONCLUSION

We showed that a competitive GINE toxicity model is *not* causally faithful by default, and that a simple causal regularizer only partially closes the gap. Reporting this honestly—with verified metrics, confidence intervals, and documented tooling limits—is more useful than an unverified faithfulness headline.

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